I Gram - Schmidt Process

Problem :

Given : $T = \{p_1 \; p_2 \; \cdots \; p_n\}$ (not necesarilly orthoronmal

Find : A set of vectors

$$T' = \{q_1 \; q_2 \; \ldots \; q_n'\}$$

that is orthonormal

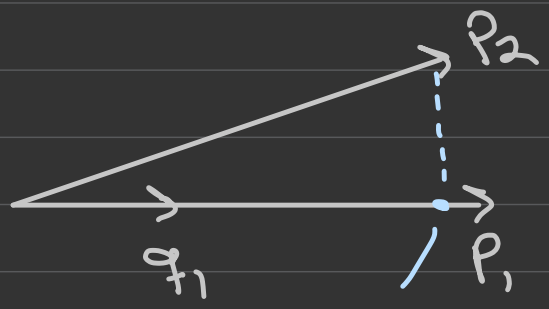
$$\langle q_i , q_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

and $\text{span}(T) = \text{span}(T')$

A) Algorithm

1) $q_1 = \dfrac{p_1}{\|p_1\|}$

2)



$\text{Proj}(p_2, q_1)$

$e_2 = p_2 - \text{proj}\langle p_2, q_1 \rangle$

$e_2 = p_2 - \dfrac{\langle p_2, q_1 \rangle}{\|q_1\|^2} q_1$

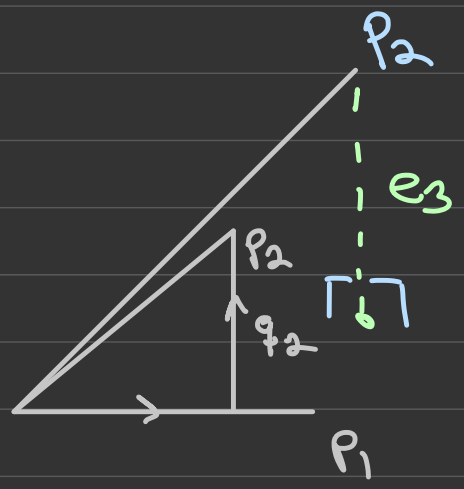
$e_2 = p_2 - \langle p_2, q_1 \rangle q_1$

q_1 has already been normalized $\|q_1\|^2 = 1$

Now normalize q_2

$$q_2 = \frac{e_2}{\|e_2\|}$$

3)



$e_3 = p_3 - \text{proj}(p_3, \{q_1\ q_2\})$

e_3 must be orthonormal to q_1 and q_2

That is like writing

$$\hat{p}_3 = a_1 q_1 + a_2 q_2$$

And we want to minimize

$$\min \|p_3 - \hat{p}_3\|_2^2$$

Thus we say that we want the error to be orthogonal to the "data". The data in this case is q_1 and q_2. Thus we want

$$p_3 - (a_1 q_1 + a_2 q_2) \perp q_1, q_2$$

To find that we write

$$\langle p_3 - (a_1 q_1 + a_2 q_2), q_1 \rangle = 0$$

$$\langle p_3 - (a_1 q_1 + a_2 q_2), q_2 \rangle = 0$$

FOIL out and solve for a_1 and a_2

$$\begin{bmatrix} \langle q_1, q_1 \rangle & \langle q_2, q_1 \rangle \\ \langle q_1, q_2 \rangle & \langle q_2, q_2 \rangle \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \langle p_3 \, q_1 \rangle \\ \langle p_3 \, q_2 \rangle \end{bmatrix}$$

Because we normalized the vectors and made them orthogonal, we simplify to

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{matrix} \langle p_3, q_1 \rangle \\ \langle p_3, q_2 \rangle \end{matrix}$$

$$a_1 = \langle p_3, q_1 \rangle \qquad a_2 = \langle p_3, q_2 \rangle$$

Now going back we can say

$$e_3 = p_3 - (\langle p_3, q_1 \rangle q_1 + \langle p_3, q_2 \rangle q_2)$$

And we guaranteed $e_3 \perp q_1$ and $e_3 \perp q_2$

And normalize $q_3 = \dfrac{e_3}{\|e_3\|}$

4) For the $k = 1, 2, \dots, n^{th}$ step

$$e_k = p_k - \text{Proj}(p_k, \{q_1, q_2, \dots, q_k\})$$

$$= p_k - \sum_{i=1}^{k-1} \langle p_k, q_1 \rangle q_1$$

$$q_k = \frac{e_k}{\|e_k\|}$$

A) Matrix Form

Let $A = \begin{bmatrix} | & | & & | \\ p_1 & p_2 & \dots & p_n \\ | & | & & | \end{bmatrix}$

Note that $p_k = e_k + \sum_{i=1}^{k-1} \langle p_k, q_i \rangle q_i$

also $e_k = q_k \|e_k\|$

$$[p_1 \ p_2 \ \cdots \ p_n] = [q_1 \ q_2 \ \cdots \ q_n] \begin{bmatrix} \|q_1\| & \langle p_2, q_1 \rangle & \cdots \\ 0 & \|e_1\| & \langle p_3, q_2 \rangle \\ 0 & 0 & \\ \vdots & 0 & \cdots \\ \vdots & \vdots & \cdots \end{bmatrix}$$

More simply

$$A = QR$$

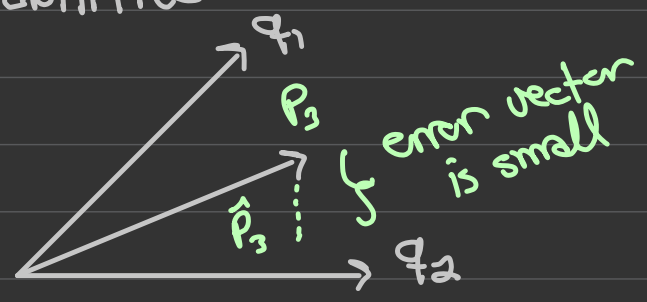
$$A = [\, p_1 \;\; p_2 \;\; \cdots \;\; p_n \,]$$

$$Q = [\, q_1 \;\; q_2 \;\; \cdots \;\; q_n \,]$$

$$R = \begin{bmatrix} \|p_1\| & \langle p_2, q_1 \rangle & \langle p_3, q_1 \rangle & \cdots \\ 0 & \|e_2\| & \langle p_3, q_2 \rangle & \cdots \\ 0 & 0 & \|e_3\| & \cdots \\ 0 & 0 & 0 & \cdots \end{bmatrix}$$

B) Vulnerabilities

Consider



The closer p_3 gets to being linear dependent on q_1 and q_2, the more we don't really need p_3. In other words, its gets so small that numerical precision matters!

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